Arista Access Node Hardware/Environmental Fault Resolution Procedure for Whitefield POP

Node: WFD-ACC-48 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_8590

Description

The Arista access node, situated at the Whitefield POP, has encountered a hardware/environmental fault, specifically manifesting as ASIC memory parity errors, line card crashes, or power supply redundancy loss, culminating in the generation of ALARM_8590 with a minor severity. This failure domain is precipitated by the intricate interplay of hardware components and environmental factors, including the ASIC's memory parity checking mechanism, which detects and reports discrepancies in data storage, and the line cards' susceptibility to thermal fluctuations and power supply anomalies. The Arista access node's role in the network topology, facilitating connectivity and data transmission between various network segments, renders it vulnerable to these types of faults. Furthermore, the equipment's design and configuration, including the implementation of redundant power supplies and line cards, can also contribute to the occurrence of these faults. Understanding the underlying hardware mechanics and protocol interactions is essential for devising an effective resolution strategy. The Arista CLI dialect will be employed to execute diagnostic and resolution commands, ensuring compatibility and efficacy.

Pre-requisite Checks

- Dispatch remote hands to Whitefield POP to perform physical checks

- Verify redundant link status to ensure no single points of failure

- Check environmental conditions, such as temperature and humidity, to ensure they are within the specified ranges

Potential Root Causes

Cause	Probability
ASIC memory parity errors due to hardware degradation	Medium
Line card crashes resulting from power supply anomalies	High
Power supply redundancy loss caused by faulty components	Low

Diagnostic Commands & Expected Outputs

```
Command: show hardware acl-asic errors, Expected Output: List of ASIC memory parity errors, Description: This command isolates ASIC memory parity errors, providing insight into the hardware's integrity and facilitating the identification of faulty components.

Command: show interfaces transceiver details, Expected Output: Transceiver information, including temperature and voltage,
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Description: This command provides detailed information about the transceivers, enabling the identification of potential issues related to temperature, voltage, or other environmental factors.

Command: show power-supply, Expected Output: Power supply status, including redundancy and voltage, Description: This command displays the power supply status, allowing for the verification of redundancy and voltage levels, which is crucial for identifying power-related issues.
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Resolution Steps

Step 1: Step: Replace faulty ASIC, Command: hardware module replace asic , Description: This step involves replacing the faulty ASIC, which requires careful planning and execution to minimize network downtime. The 'hardware module replace asic' command is used to replace the faulty ASIC, ensuring that the new component is properly integrated into the system.

Step 2: Step: Reset line card, Command: hardware module reset , Description: This step involves resetting the line card, which can help resolve issues related to power supply anomalies or other environmental factors. The 'hardware module reset' command is used to reset the line card, ensuring that it is properly initialized and functioning correctly.

Step 3: Step: Replace power supply, Command: hardware power-supply replace , Description: This step involves replacing the faulty power supply, which is crucial for maintaining redundancy and ensuring the overall reliability of the system. The 'hardware power-supply replace' command is used to replace the faulty power supply, ensuring that the new component is properly integrated into the system.

Post-Check Verification

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Command: show hardware acl-asic errors, Expected Output: No ASIC memory parity errors, Description: This command verifies that the ASIC memory parity errors have been resolved, ensuring that the hardware is functioning correctly.

Command: show interfaces transceiver details, Expected Output: Transceiver information, including temperature and voltage, within normal ranges, Description: This command verifies that the transceivers are functioning correctly, with temperature and voltage levels within the specified ranges.

Command: show power-supply, Expected Output: Power supply status, including redundancy and voltage, within normal ranges, Description: This command verifies that the power supply status is normal, with redundancy and voltage levels within the specified ranges.
```

Roll-back Procedures

- Step: Revert to previous ASIC configuration, Command: hardware module configure asic default, Description: This step involves reverting to the previous ASIC configuration, which can help resolve issues related to the replacement of the faulty ASIC. The 'hardware module configure asic' command is used to revert to the default configuration, ensuring that the system is restored to its previous state.

- Step: Revert to previous line card configuration, Command: hardware module configure default, Description: This step involves reverting to the previous line card configuration, which can help resolve issues related to the reset of the line card. The 'hardware module configure' command is used to revert to the default configuration, ensuring that the system is restored to its previous state.

- Step: Revert to previous power supply configuration, Command: hardware power-supply configure default, Description: This step involves reverting to the previous power supply configuration, which can help resolve issues related to the replacement of the faulty power supply. The 'hardware power-supply

configure' command is used to revert to the default configuration, ensuring that the system is restored to its previous state.